

SCRIPT-CONTROLLED ACL

Restrict Record Access Using Field Values

ServiceNow Micro Project Documentation

Team ID: SWTID-2026-2956

Team Size: 5

Team Leader: D.srinithi

Team member: Kanimozhi

Team member: Rajalakshmi

Team member: Seetha. A

Team member: Arivoli R

1. Project Overview

This practical project demonstrates how a script-controlled Access Control List (ACL) can be used in ServiceNow to control record access according to a field value. The example uses the Institution Details table and the Branch field as the basis for access control. Users associated with the EEE branch are permitted to work with EEE records, while administrators retain unrestricted access.

2. Milestone 1 – User and Role Setup

2.1 Create the Test User

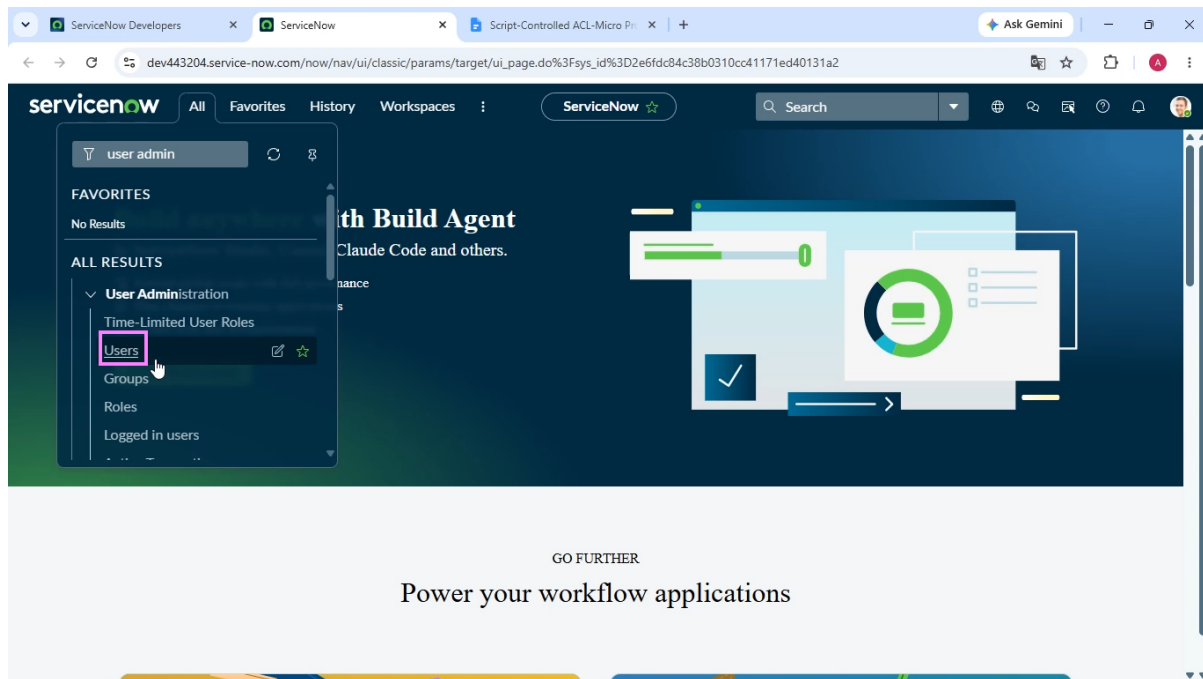
1. Sign in to the ServiceNow instance with an account that has administrator privileges.
2. Navigate to User Administration → Users.
3. Select New.
4. Create the user with the following information:

Field	Value
User ID	EEE User
First Name	EEE
Last Name	User
Email	eeeuser@gmail.com

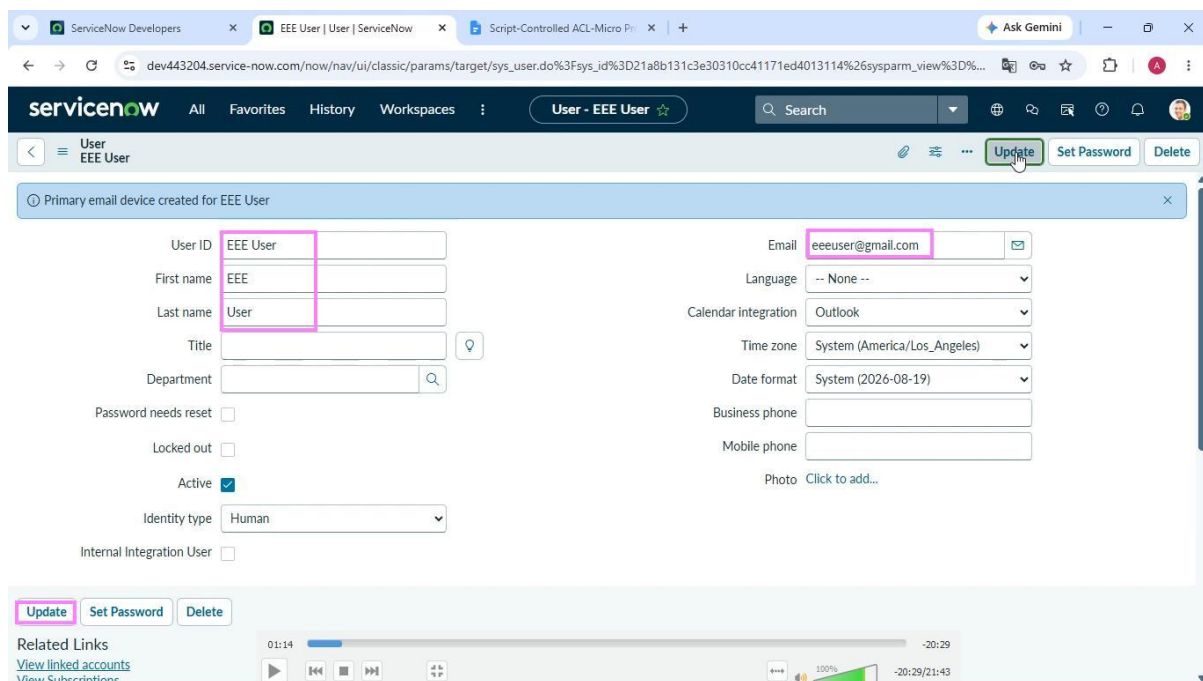
5. Save the user record.

2.2 Create the Required Roles

6. Open User Administration → Roles.
7. Select New.
8. Create the role bb1 and provide a short description explaining its purpose.
9. Select Submit.
10. Repeat the same process to create bb2, bb3, and bb4.
11. Assign all four roles (bb1, bb2, bb3, and bb4) to the EEE User.



Role and user configuration in ServiceNow



Role assignment view

3. Milestone 2 – Build the Custom Table

12. From the Application Navigator, open System Definition → Tables.
13. Select New and create the custom table with the following values:

Property	Configuration
Label	Institution Details
Name	u_institution_details

Extends	False
---------	-------

14. Submit the table.

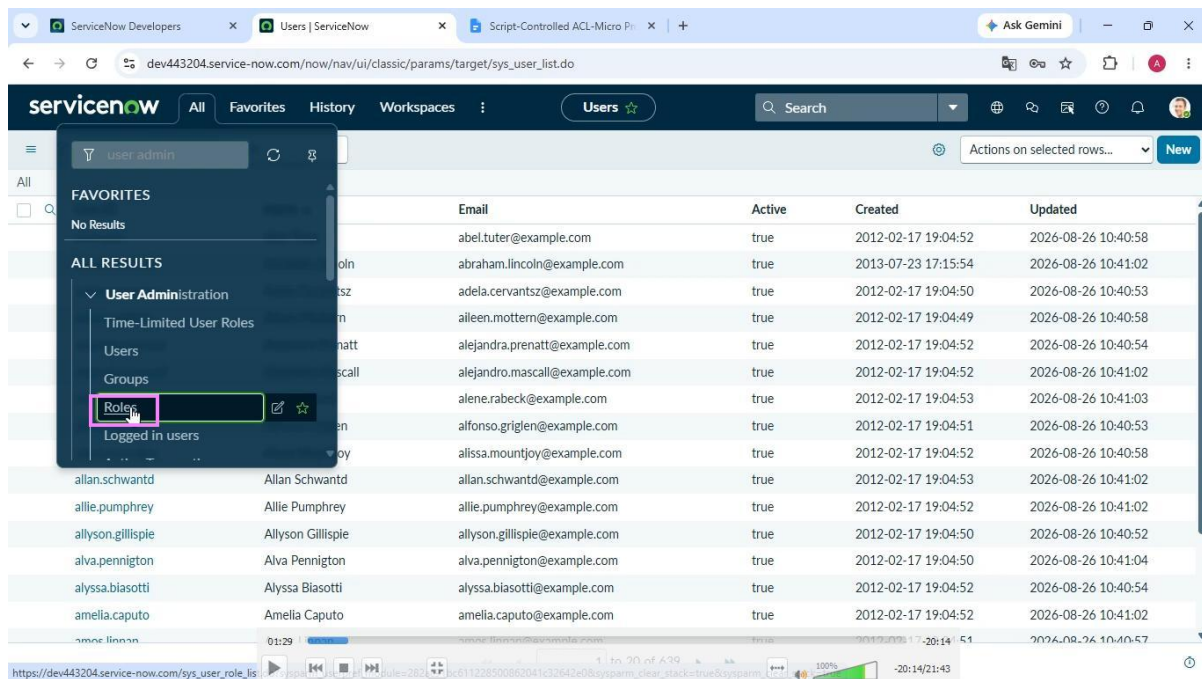
3.1 Configure the Table Fields

Add the following fields to u_institution_details:

Field	Type	Values / Reference
Student Roll Number	Auto Number	—
Student Name	Reference	User
Faculty Name	Reference	User
Branch	Choice	ECE, EEE, CSE
Email	String	—
Phone Number	String	—
Description	Multi String	—

15. Save the table configuration.

16. Create several sample records and use different Branch values such as ECE, EEE, and CSE.



Custom table configuration

ServiceNow Developers | New Record | Role | ServiceNow | Script-Controlled ACL-Micro Pr | Ask Gemini

dev443204.service-now.com/now/nav/ui/classic/params/target/sys_user_role.do%3Fsys_id%3D1%26sys_is_list%3Dtrue%26sys_target%3Dsys_user_role%2...

servicenow All Favorites History Workspaces Role - New Record Search

Role New record

Name Application Global Elevated privilege ☐

Description

Submit

01:43 -20:00

Table field configuration

ServiceNow Developers

Users | ServiceNow

Script-Controlled ACL-Micro Pr

Ask Gemini

dev443204.service-now.com/now/nav/ui/classic/params/target/sys_user_list.do%3Fsysparm_first_row%3D1%26sysparm_query%3D%3DGOTOName%253E%25...

🔍 ⌂ 🔄 ⌚ 📧

servicenow

All Favorites History Workspaces ⋮ Users ⭐

🔍 Search

🌐 🔄 📄 ⌚ 📧

Actions on selected rows... New

🔍 system def

FAVORITES

No Results

ALL RESULTS

System Definition

- Record Hierarchy
- Application Menus
- Menu Categories
- Dictionary
- Tables

Email	Active	Created	Updated		
Search	Search	Search	Search		
eeeuser@gmail.com	true	2026-09-26 01:10:27	2026-09-26 01:10:27		
efren.baucher@example.com	true	2012-02-17 19:04:53	2026-08-26 10:40:57		
elaine.renzi@example.com	true	2012-02-17 19:04:52	2026-08-26 10:40:54		
eldon.sutch@example.com	true	2012-02-17 19:04:53	2026-08-26 10:41:06		
eli.bettner@example.com	true	2012-02-17 19:04:51	2026-08-26 10:40:57		
elisa.gracely@example.com	true	2012-02-17 19:04:52	2026-08-26 10:40:54		
eliseo.wice@example.com	true	2012-02-17 19:04:53	2026-08-26 10:40:54		
ella.pahnke@example.com	true	2012-02-17 19:04:49	2026-08-26 10:41:03		
elmo.dagenais	Elmo Dagenais	elmo.dagenais@example.com	true	2012-02-17 19:04:53	2026-08-26 10:41:04
elmo.gabouer	Elmo Gabouer	elmo.gabouer@example.com	true	2012-02-17 19:04:53	2026-08-26 10:40:52
elvira.blumenthal	Elvira Blumenthal	elvira.blumenthal@example.com	true	2012-02-17 19:04:52	2026-08-26 10:41:02
emery.reek	Emery Reek	emery.reek@example.com	true	2012-02-17 19:04:49	2026-08-26 10:40:55
emilia.oxley	Emilia Oxley	emilia.oxley@example.com	true	2012-02-17 19:04:49	2026-08-26 10:41:02
emilio.lampkin	Emilio Lampkin	emilio.lampkin@example.com	true	2012-02-17 19:04:49	2026-08-26 10:41:06

04:10

1 to 20 of 460

100%

-17:33/21:43

Sample Institution Details records

Additional table configuration view

4. Milestone 3 – Configure the READ ACL

17. Open System Security → Access Control (ACL) from the Application Navigator.
18. Elevate the security_admin role.
19. Select New to create the ACL.
20. Configure the ACL as follows:

ACL Property	Value
Type	record
Operation	read
Name	u_institution_details
Active	true
Advanced	true

21. In the Requires role related list, add the bb1 role.
22. Set the data condition so that Branch is EEE.
23. Enter the following script in the Script field:

```
(function () {
    // Administrators are permitted to access every record.
    if (gs.hasRole('admin')) {
        return true;
    }

    // Permit users with the bb1 role to access the EEE records.
    if (gs.hasRole('bb1')) {
        return true;
    }

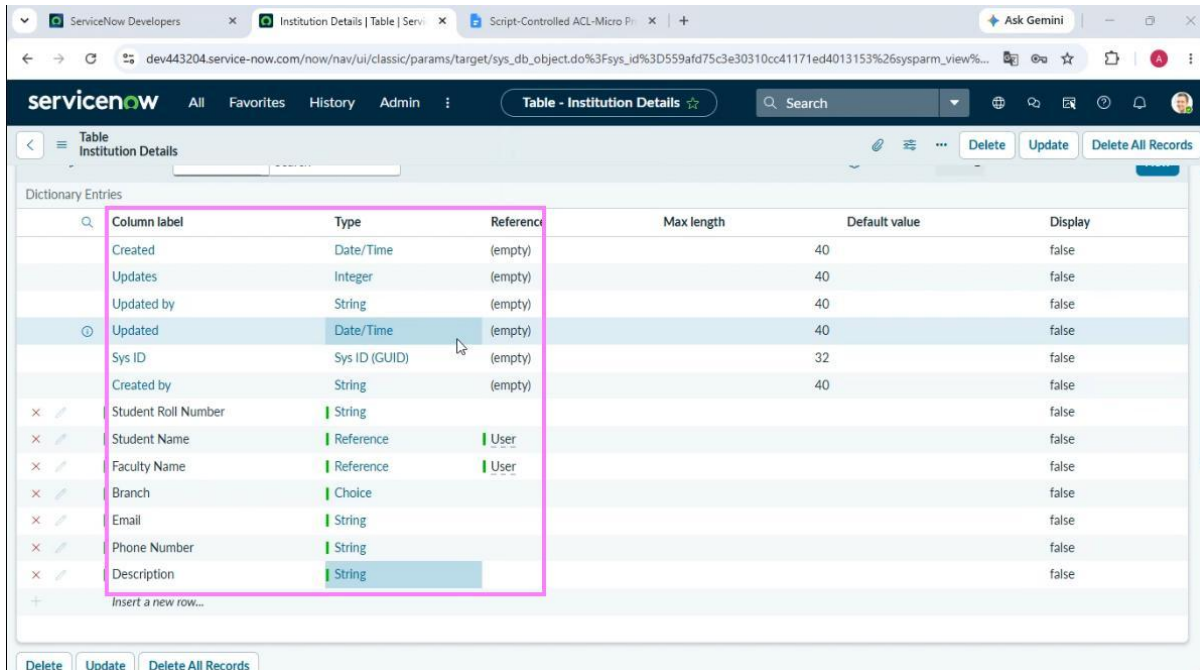
    // Reject access for users who do not satisfy the rule.
```

```

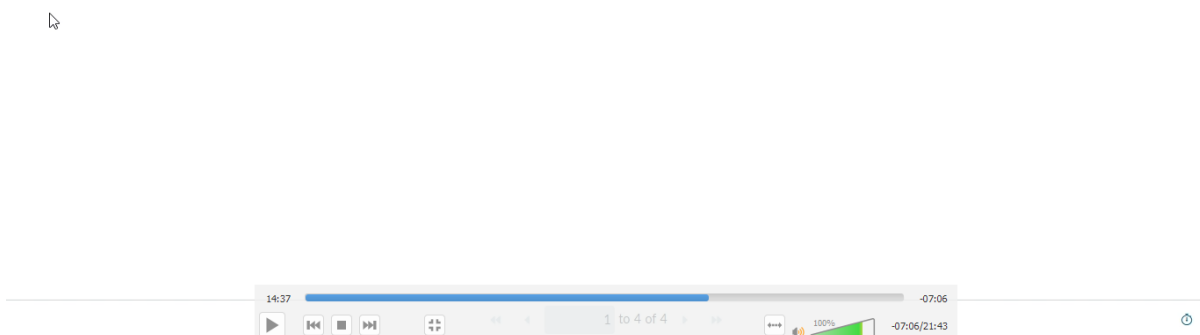
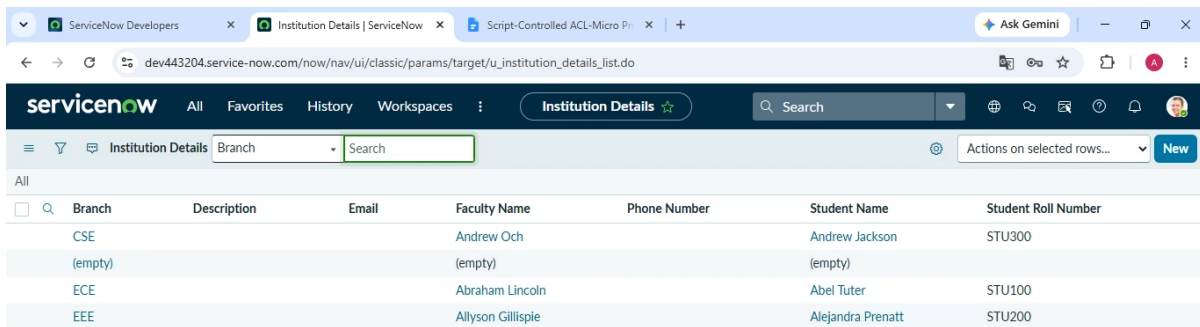
return false;
})();

```

24. Select Submit to save the ACL.



READ ACL configuration



READ ACL script and condition

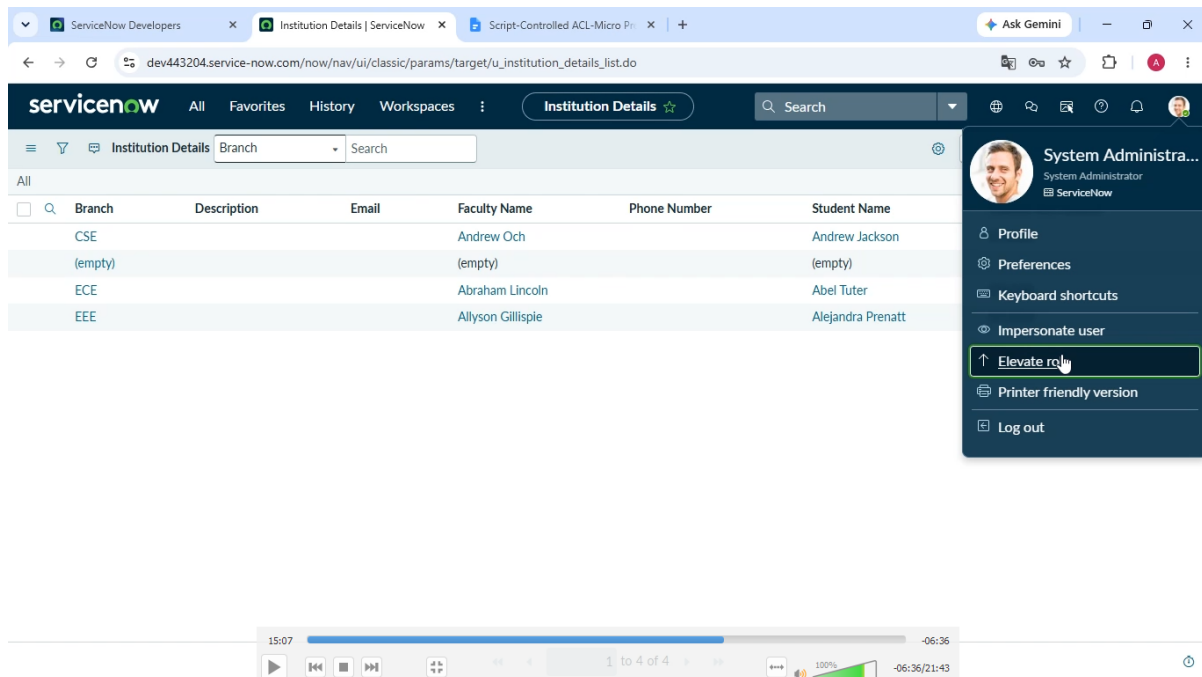
4.1 READ ACL Validation

25. Impersonate the EEE User and open the Student Records list.
26. Use u_institution_details.list (or LIST) in the All menu to open the table list.
27. Impersonate the Admin user and open the same list.

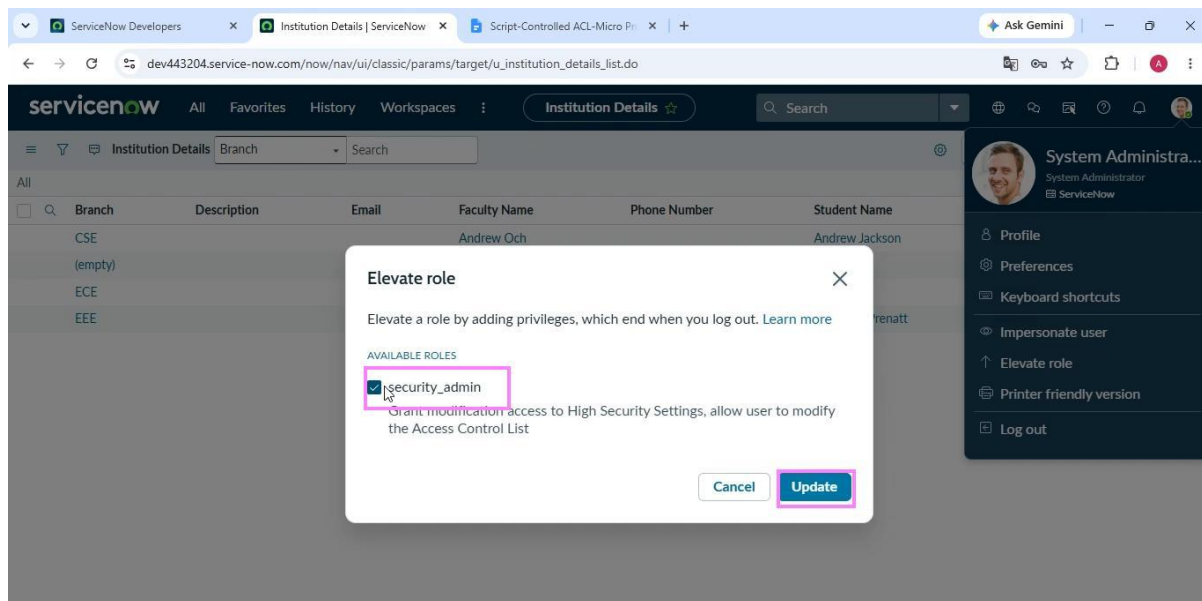
The expected behavior is:

Script-Controlled ACL – Restrict Record Access Using Field Values

- A user with the bb1 role can access the EEE branch records.
- A user without the required role cannot access the protected records.
- An administrator can access all records regardless of the Branch value.



ACL list and access configuration



READ ACL validation

5. Milestone 4 – Configure the CREATE ACL

28. Select New to create another ACL.
29. Set Type to record.
30. Set Operation to create.
31. Set Name to u_institution_details.
32. Keep Active enabled.

Script-Controlled ACL – Restrict Record Access Using Field Values

33. Add bb2 to the Requires role related list.
34. Leave the Data condition empty.
35. Save the ACL.

5.1 CREATE ACL Validation

A user who has both bb1 and bb2 can access the EEE records and receives the New button for creating records.

The screenshot displays the 'Access Control' configuration page for the resource 'u_institution_details'. The configuration includes the following elements:

- Type:** record
- Operation:** read
- Decision Type:** Allow If
- Admin overrides:** ☒
- Protection policy:** -- None --
- Name:** Institution Details [u_institution_details]
- Description:** (empty text area)
- Applies To:** No. of records matching the condition: 4
- Data Condition:** (empty)
- Advanced Condition:** A script editor containing the following code:


```

1  (function() {
2    if (gs.hasRole('bb1')) {
3      return true;
4    } else {
5      return false;
6    }
7  })();
      
```

CREATE ACL configuration and validation

6. Milestone 5 – Configure the WRITE ACL

36. Select New to create the Write ACL.
37. Set Type to record.
38. Set Operation to write.
39. Set Name to u_institution_details.
40. Keep Active enabled.
41. Add bb3 to the Requires role related list.
42. Leave the Data condition unconfigured.
43. Save the ACL.

6.1 WRITE ACL Validation

A user with bb1, bb2, and bb3 can view and modify the EEE branch records and can also use the New button.

Access Control
u_institution_details

* Type: record

* Operation: create

Decision Type: Allow If

Admin overrides: ☒

Protection policy: -- None --

* Name: Institution Details (u_institution_details)

Description:

Applies To: No. of records matching the condition: 6

Add Filter Condition Add OR Clause

-- choose field -- -- op -- -- value --

Application: Global

Active: ☒

Advanced: ☐

Update Delete

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.

2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role
BB2
Insert a new row...

1 to 1 of 1

WRITE ACL configuration and validation

7. Milestone 6 – Configure the DELETE ACL

44. Select New to create the Delete ACL.
45. Set Type to record.
46. Set Operation to delete.
47. Set Name to u_institution_details.
48. Keep Active enabled.
49. Add bb4 to the Requires role related list.
50. Leave the Data condition empty.
51. Save the ACL.

7.1 DELETE ACL Validation

A user with all four roles—bb1, bb2, bb3, and bb4—can view, create, edit, and delete the EEE branch records.

Access Control
u_institution_details

* Type: record

* Operation: write

Decision Type: Allow if

Admin overrides: ☒

Protection policy: -- None --

* Name: Institution Details [u_institution_details] -- None --

Description:

Applies To: No. of records matching the condition: 6

Add Filter Condition Add OR Clause

-- choose field -- -- oper -- -- value --

Application: Global

Active: ☒

Advanced: ☐

Update Delete

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role

BB3

Insert a new row...

1 to 1 of 1

DELETE ACL configuration

Access Control
u_institution_details

* Type: record

* Operation: write

Decision Type: Allow if

Admin overrides: ☒

Protection policy: -- None --

* Name: Institution Details [u_institution_details] -- None --

Description:

Applies To: No. of records matching the condition: 6

Add Filter Condition Add OR Clause

-- choose field -- -- oper -- -- value --

Application: Global

Active: ☒

Advanced: ☐

Update Delete

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role

BB3

Insert a new row...

1 to 1 of 1

DELETE ACL validation

8. Outcome

The completed ACL configuration provides record-level security for the Institution Details table. Access is determined through the combination of user roles, ACL operations, the Branch field condition, and the advanced security script.

The permissions are separated across READ, CREATE, WRITE, and DELETE operations. This structure limits protected records to authorized users while preserving unrestricted access for administrators.

Demo: <https://drive.google.com/file/d/1LaGSNAZQP5fRwoYEI3iyNDJZv73I-qCF/view?usp=drivesdk>